

# Homework 1

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## Abstract

This is my abstract!

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### 1. QUESTION 1

1.a. *Proof*

Start by taking logs on both sides of the return expression:

$$\log R_{e,t+1} = \log(1 + \exp(\log D_{t+1} - \log P_{t+1})) + \log P_{t+1} - \log P_t \quad (1)$$

For simplicity, I denote  $r_{e,t+1} = \log R_{e,t+1}$ ,  $p_{t+1} = \log P_{t+1}$ ,  $d_{t+1} = \log D_{t+1}$ , and  $dp_{t+1} = \log D_{t+1} - \log P_{t+1}$ , then, by a Taylor expansion in  $dp_{t+1}$  (which is stationary with mean  $\bar{dp}$ ), we get

$$r_{e,t+1} = \log(1 + \exp(\bar{dp})) + \frac{\exp(\bar{dp})}{1 + \exp(\bar{dp})} (dp_{t+1} - \bar{dp}) + p_{t+1} - p_t. \quad (2)$$

Denote  $\kappa = 1/(1 + \exp(\bar{dp}))$ , and  $\kappa_0 = \log(1 + \exp(\bar{dp})) - \bar{dp} \exp(\bar{dp})/(1 + \exp(\bar{dp})) = -\log \kappa - (1 - \kappa) \log(1/\kappa - 1)$ . The log-linearization becomes

$$r_{e,t+1} = \kappa_0 + (1 - \kappa) dp_{t+1} + p_{t+1} - p_t. \quad (3)$$

Moving  $p_t$  to the left-hand side and the other terms to the right-hand side, we get

$$\begin{aligned} p_t &= \kappa_0 + (1 - \kappa) d_{t+1} - r_{e,t+1} + \kappa p_{t+1} \\ &= \kappa_0 + (1 - \kappa) d_{t+1} - r_{e,t+1} + \kappa [\kappa_0 + (1 - \kappa) d_{t+2} - r_{e,t+2} + \kappa p_{t+2}] \\ &= \kappa_0 (1 + \kappa + \dots + \kappa^{H-1}) \\ &\quad + (1 - \kappa) (d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1} d_{t+H}) \\ &\quad - [r_{e,t+1} + \kappa r_{e,t+2} + \dots + \kappa^{H-1} r_{e,t+H}] \\ &\quad + \kappa^H p_{t+H} \end{aligned} \quad (4)$$

The log dividend-price ratio  $dp_t$  can then be written as

$$\begin{aligned}
dp_t &= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\
&\quad + [r_{e,t+1} + \kappa r_{e,t+1} + \dots + \kappa^{H-1} r_{e,t+H}] \\
&\quad d_t - (1 - \kappa)(d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1} d_{t+H}) \\
&\quad - \kappa^H p_{t+H} \\
&= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\
&\quad + [r_{e,t+1} + \kappa r_{e,t+1} + \dots + \kappa^{H-1} r_{e,t+H}] \\
&\quad d_t - d_{t+1} + \kappa(d_{t+1} - d_{t+2}) + \dots + \kappa^{H-1}(d_{t+H-1} - d_{t+H}) \\
&\quad + \kappa^H d_{t+H} - \kappa^H p_{t+H} \\
&= \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}
\end{aligned} \tag{5}$$

where  $\Delta d_{t+1} := d_{t+1} - d_t$ . Notice that the identity above holds ex post (state by state), so it also holds in conditional expectation at time  $t$ , giving

$$dp_t = \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} + \kappa^H \mathbb{E}_t dp_{t+H} \tag{6}$$

Letting  $H \rightarrow \infty$  and imposing the no-bubble (transversality) condition

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t dp_{t+H} = 0 \tag{7}$$

we obtain equation (1.3):

$$dp_t = \kappa_0 \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \tag{8}$$

1.b.

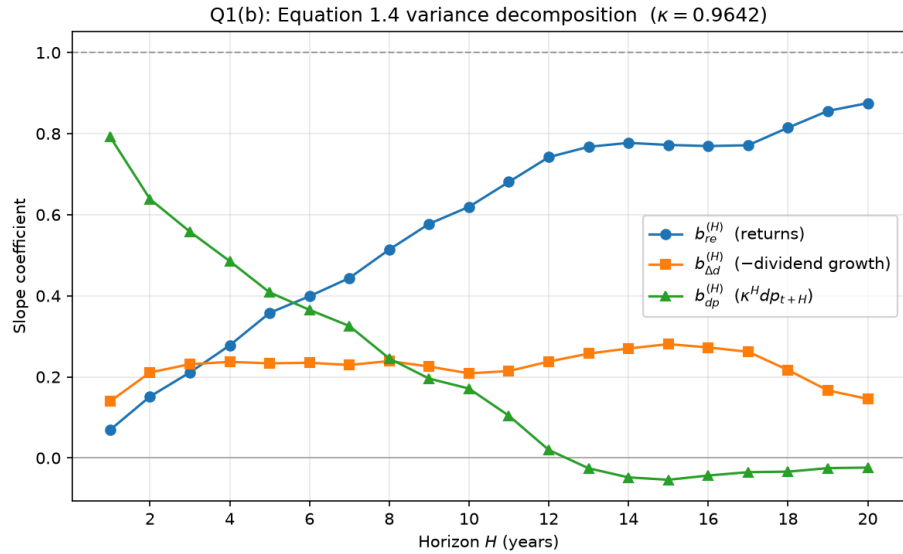


Figure 1: Equation 1.4 variance-decomposition slopes vs. horizon  $H$  (EQ Dataset, 1928–2021; overlapping monthly starts;  $\kappa = 1/(1 + e^{\overline{dp}}) = 0.9642$ ).

1.c.

1.d. *Alternative log-linear present-value identity*

Using the same trick,

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{D_{t+1}} \frac{D_{t+1}}{D_t} \frac{D_t}{P_t + D_t} \frac{P_t + D_t}{P_t} \quad (9)$$

, the log one-period return can be written as

$$r_{e,t+1} = -\log(1 - e^{-dy_{t+1}}) + \Delta d_{t+1} + \log(1 - e^{-dy_t}) + dy_t \quad (10)$$

where  $dy_t := \log\left(1 + \frac{D_t}{P_t}\right)$  and  $\log(1 - e^{-dy_t}) = \log \frac{D_t}{P_t + D_t}$ .

Applying a Taylor expansion of  $\log(1 - e^{-dy_t})$  in  $dy_t$  around its mean  $\bar{dy}$ , we have

$$\log(1 - e^{-dy_t}) = \log(1 - e^{-\bar{dy}}) + \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}} (dy_t - \bar{dy}) + O\left((dy_t - \bar{dy})^2\right) \quad (11)$$

Letting  $\kappa = e^{-\bar{dy}}$  to simplify the expression,

$$\log(1 - e^{-dy_t}) = \log(1 - \kappa) + \frac{\kappa}{1 - \kappa} (dy_t - \bar{dy}) + O\left((dy_t - \bar{dy})^2\right) \quad (12)$$

Then the return is log-linearized as

$$r_{e,t+1} = -\frac{\kappa}{1 - \kappa} (dy_{t+1} - \bar{dy}) + \Delta d_{t+1} + \frac{\kappa}{1 - \kappa} (dy_t - \bar{dy}) + dy_t \quad (13)$$

and

$$\begin{aligned} \frac{dy_t}{1 - \kappa} &= r_{e,t+1} - \Delta d_{t+1} + \kappa \frac{dy_{t+1}}{1 - \kappa} \\ &= r_{e,t+1} - \Delta d_{t+1} \\ &\quad + \kappa \left[ r_{e,t+2} - \Delta d_{t+2} + \kappa \frac{dy_{t+2}}{1 - \kappa} \right] \\ &= r_{e,t+1} - \Delta d_{t+1} + \kappa [r_{e,t+2} - \Delta d_{t+2}] \\ &\quad + \dots + \kappa^{H-1} [r_{e,t+H} - \Delta d_{t+H}] + \kappa^H \frac{dy_{t+H}}{1 - \kappa} \\ &= \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H \frac{dy_{t+H}}{1 - \kappa} \end{aligned} \quad (14)$$

Thus, the log-linearization identity is

$$dy_t = (1 - \kappa) \left( \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H} \quad (15)$$

Taking the conditional expectation at time  $t$ , the identity also holds

$$dy_t = (1 - \kappa) \left( \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \right) + \kappa^H \mathbb{E}_t dy_{t+H} \quad (16)$$

Imposing the no-bubble assumption, the last term vanishes as  $h \rightarrow \infty$ ,

$$\lim_{H \rightarrow \infty} \kappa^H E_t \log \left( \frac{P_t + D_t}{P_t} \right) \rightarrow 0 \quad (17)$$

## 2. QUESTION 2